1. Consider following array

int p[3][3]={1,2,3,4,5,6,7,8,9};

Assume the base address of array p=1000.

find the address of P[2][1]?(2D arrays follows Row Major ordering)

Answer: p[2][1]=1000+((2*3)+1)*4=1000+24=1024

7.Write a program in C to find the row with maximum number of 1s using Functions

The given 2D array

0 1 0 1 1

1 1 1 1 1

1 0 0 1 0

0 0 0 0 0

1 0 0 0 1

Ans:
```
#include <stdio.h>
#define R 5
#define C 5
 int getFirstOccur(int arr1[], int l, int h)
{
if(h >= l)
{
   int mid = l + (h - l)/2;
   if ( ( mid == 0 || arr1[mid-1] == 0) && arr1[mid] == 1)
   return mid;
   else if (arr1[mid] == 0)
   return getFirstOccur(arr1, (mid + 1), h);
   else
   return getFirstOccur(arr1, l, (mid -1));
```

```
}
return -1;
}
int findRowMaxOne(int arr2d[R][C])
{
    int max_row_index = 0, max = -1;
    int i, index;
    for (i = 0; i < R; i++)
    {
    index = getFirstOccur (arr2d[i], 0, C-1);
    if (index != -1 && C-index > max)
    {
        max = C - index;
        max_row_index = i;
    }
    }
    return max_row_index;
}

int main()
{
    int arr2d[R][C] = { {0, 1, 0, 1,1},
                        {1, 1, 1, 1, 1},
                        {1, 0, 0, 1, 0},
                        {0, 0, 0, 0, 0},
                        {1, 0, 0, 0, 1}
    };
    int i,j;
```

```
  printf("The given 2D array is :  \n");
        for(i = 0; i < R; i++)
        {
        for(j=0; j<C ; j++)
        {
        printf("%d  ", arr2d[i][j]);
        }
printf("\n");
   }
 printf("The index of row with maximum 1s is:  %d " , findRowMaxOne(arr2d));
   return 0;
}
```